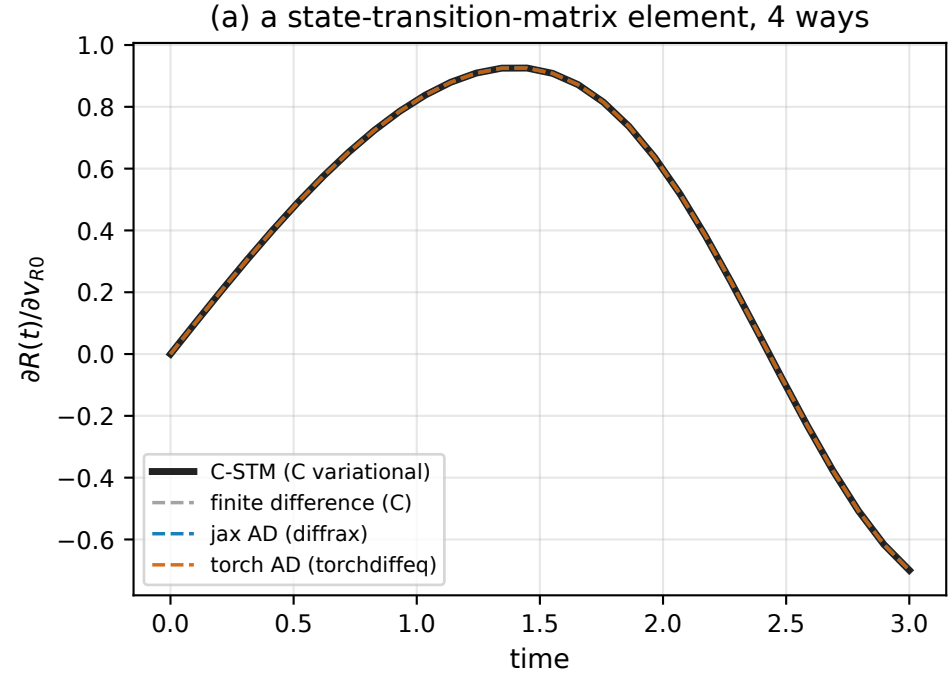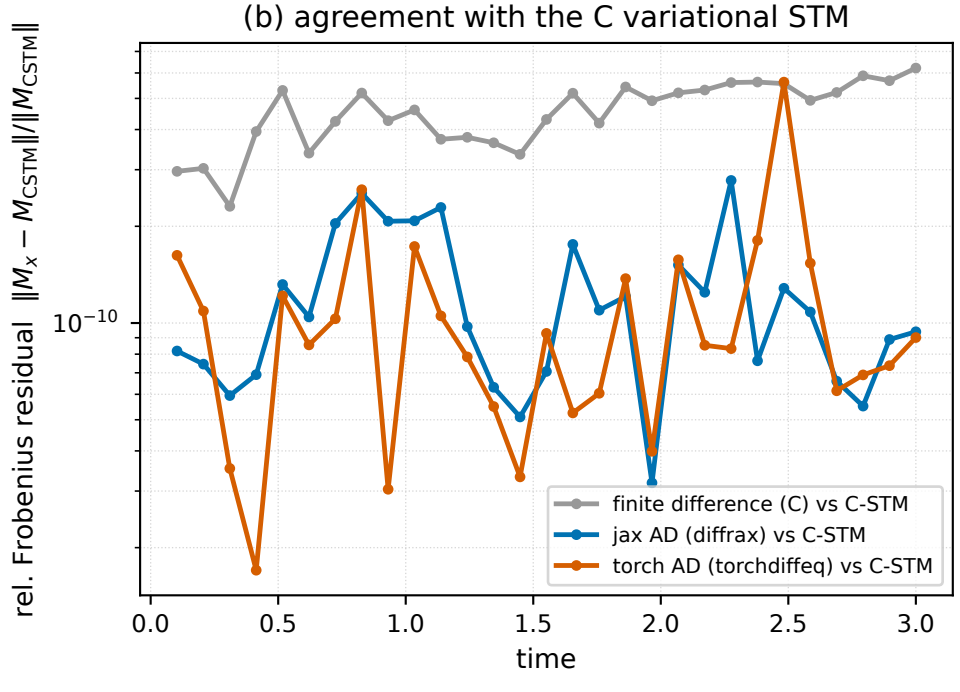$\partial x(t)/\partial x_0$: autodiff = the C state-transition matrix to solver precision (MWPotential2014)